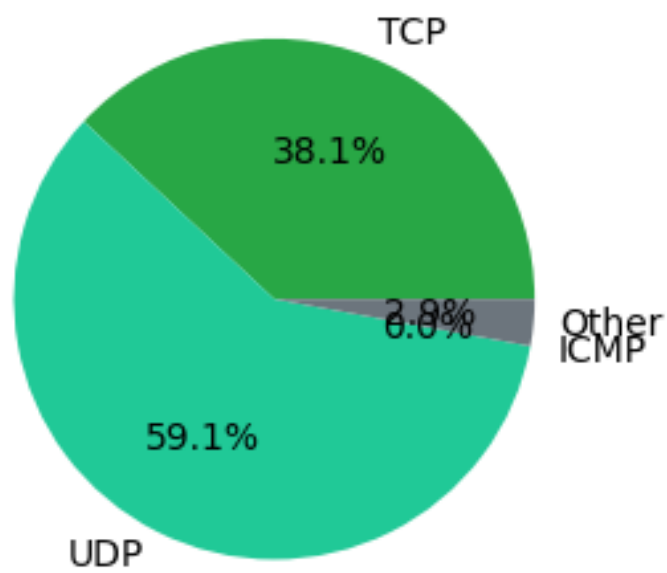


NIDS Session Report

AI-Based Network Intrusion Detection System

Protocol Distribution

Protocol Distribution



Session Summary

Start Time:	1750244719.1983979
Stop Time:	2025-06-18T16:37:11.911701
Uptime:	112 seconds
Packets Processed:	381
Anomalies Detected:	22
Packet Rate:	0 packets/sec

Protocol Statistics

TCP PACKETS:	145
UDP PACKETS:	225
ICMP PACKETS:	0
OTHER PACKETS:	11

Alerts (5)

Type: ML_ANOMALY | Time: 2025-06-18T16:35:24.266177

Details: {'anomaly_score': -7.308270290257125, 'features': {'packet_count': 16, 'total_bytes': 975, 'avg_packet_length': np.float64(60.9375), 'std_packet_length': np.float64(6.832539333951909), 'min_packet_length': 42, 'max_packet_length': 75, 'avg_payload_size': np.float64(1.5), 'std_payload_size': np.float64(0.8660254037844386), 'total_payload_size': 24, 'tcp_count': 12, 'udp_count': 0, 'icmp_count': 0, 'other_protocol_count': 0, 'protocol_diversity': 1, 'tcp_ratio': 1.0, 'udp_ratio': 0.0, 'icmp_ratio': 0.0, 'other_protocol_ratio': 0.0, 'unique_src_ports': 1, 'unique_dst_ports': 12, 'total_ports_accessed': 13, 'common_port_ratio': 0.0, 'high_port_ratio': 1.0, 'avg_dst_port': np.float64(52408.5), 'std_dst_port': np.float64(5.86657196893268), 'min_dst_port': 52400, 'max_dst_port': 52420, 'unique_connections': 9, 'total_connection_attempts': 12, 'avg_packets_per_connection': 1.3333333333333333, 'connection_diversity': 0.5625, 'max_packets_per_connection': 4, 'unique_src_ips': 9, 'unique_dst_ips': 1, 'src_ip_diversity': 0.5625, 'dst_ip_diversity': 0.0625, 'private_src_ip_ratio': 0.0, 'private_dst_ip_ratio': 1.0, 'packet_rate': 3.3261311701434355, 'inter_arrival_time_mean': np.float64(0.32069290479024254), 'inter_arrival_time_std': np.float64(0.36341086328956534), 'burst_count': 7, 'burst_ratio': 0.4666666666666667, 'payload_size_mean': np.float64(1.5), 'payload_size_std': np.float64(0.8660254037844386), 'zero_payload_ratio': 0.25, 'large_payload_ratio': 0.0, 'payload_size_entropy': np.float64(0.8112781244591328), 'syn_count': 12, 'fin_count': 0, 'rst_count': 0, 'psh_count': 0, 'ack_count': 12, 'urg_count': 0, 'syn_ratio': 1.0, 'fin_ratio': 0.0, 'rst_ratio': 0.0, 'psh_ratio': 0.0, 'ack_ratio': 1.0, 'urg_ratio': 0.0}, 'timestamp': '2025-06-18T16:35:24.266162', 'packet_count': 16, 'protocol_distribution': {'tcp': 12, 'udp': 0, 'icmp': 0}, 'connection_info': {'unique_connections': 9, 'unique_src_ips': 9, 'unique_dst_ips': 1}}

Explanation: The machine learning model detected traffic that is statistically unusual compared to normal patterns.

Type: PORT_SCAN | Time: 2025-06-18T16:35:34.569723

Details: {'src_ip': '10.0.2.15', 'dst_ip': '8.8.4.4', 'dst_port': 443, 'protocol': 17}

Explanation: A port scan was detected. This means a host is rapidly probing multiple ports on a target, which is often a precursor to an attack.

Type: DDOS_INDICATOR | Time: 2025-06-18T16:36:17.101474

Details: {'src_ip': '34.198.21.75', 'dst_ip': '10.0.2.15', 'packet_size': 1494, 'protocol': 6}

Explanation: Possible Distributed Denial of Service (DDoS) activity detected. This means a large number of packets are being sent to a target, which may overwhelm the system.

Type: ML_ANOMALY | Time: 2025-06-18T16:36:24.482648

Details: {'anomaly_score': -7.296871928074141, 'features': {'packet_count': 100, 'total_bytes': 18038, 'avg_packet_length': np.float64(180.38), 'std_packet_length': np.float64(338.2605439598299), 'min_packet_length': 42, 'max_packet_length': 1536, 'avg_payload_size': np.float64(126.71), 'std_payload_size': np.float64(339.8651289850137), 'total_payload_size': 12671, 'tcp_count': 73, 'udp_count': 23, 'icmp_count': 0, 'other_protocol_count': 0, 'protocol_diversity': 2, 'tcp_ratio': 0.7604166666666666, 'udp_ratio': 0.23958333333333334, 'icmp_ratio': 0.0, 'other_protocol_ratio': 0.0, 'unique_src_ports': 19, 'unique_dst_ports': 23, 'total_ports_accessed': 23, 'common_port_ratio': 0.3333333333333333, 'high_port_ratio': 0.6666666666666666, 'avg_dst_port': np.float64(36144.510416666664), 'std_dst_port': np.float64(25458.652260820712), 'min_dst_port': 53, 'max_dst_port': 62470, 'unique_connections': 28, 'total_connection_attempts': 96, 'avg_packets_per_connection': 3.4285714285714284,

'connection_diversity': 0.28, 'max_packets_per_connection': 15, 'unique_src_ips': 16, 'unique_dst_ips': 14, 'src_ip_diversity': 0.16, 'dst_ip_diversity': 0.14, 'private_src_ip_ratio': 0.125, 'private_dst_ip_ratio': 0.14285714285714285, 'packet_rate': 2.259879237229119, 'inter_arrival_time_mean': np.float64(0.4469712334449845), 'inter_arrival_time_std': np.float64(1.237071539198024), 'burst_count': 50, 'burst_ratio': 0.5050505050505051, 'payload_size_mean': np.float64(126.71), 'payload_size_std': np.float64(339.8651289850137), 'zero_payload_ratio': 0.18, 'large_payload_ratio': 0.06, 'payload_size_entropy': np.float64(0.9322286142916755), 'syn_count': 35, 'fin_count': 2, 'rst_count': 2, 'psh_count': 13, 'ack_count': 65, 'urg_count': 0, 'syn_ratio': 0.4794520547945205, 'fin_ratio': 0.0273972602739726, 'rst_ratio': 0.0273972602739726, 'psh_ratio': 0.1780821917808219, 'ack_ratio': 0.8904109589041096, 'urg_ratio': 0.0}, 'timestamp': '2025-06-18T16:36:24.482634', 'packet_count': 100, 'protocol_distribution': {'tcp': 73, 'udp': 23, 'icmp': 0}, 'connection_info': {'unique_connections': 28, 'unique_src_ips': 16, 'unique_dst_ips': 14}}

Explanation: The machine learning model detected traffic that is statistically unusual compared to normal patterns.

Type: PORT_SCAN | Time: 2025-06-18T16:36:40.205137

Details: {'src_ip': '10.0.2.15', 'dst_ip': '142.250.77.100', 'dst_port': 443, 'protocol': 17}

Explanation: A port scan was detected. This means a host is rapidly probing multiple ports on a target, which is often a precursor to an attack.

Glossary of Terms

Packets Processed:	Total number of network packets analyzed by the NIDS during this session.
Anomalies Detected:	Number of network events flagged as suspicious or abnormal by the NIDS.
Packet Rate:	Current rate at which packets are being processed.
Uptime:	How long the NIDS was running.
TCP/UDP/ICMP/Other:	Number of packets for each protocol.
port_scan:	A port scan was detected. This means a host is rapidly probing multiple ports on a target, which is often a precursor to an attack.
ddos_indicator:	Possible Distributed Denial of Service (DDoS) activity detected. This means a large number of packets are being sent to a target, which may overwhelm the system.
unusual_protocol:	Unusual network protocol detected. This could indicate suspicious or unauthorized network activity.
ml_anomaly:	The machine learning model detected traffic that is statistically unusual compared to normal patterns.